Where did the tiny bits go?

Electrons

Name _____ Date _____



Little Wonderer studied an atom model under the lamp. An atom is a tiny building block of matter. Its nucleus has protons and neutrons packed in the middle. Electrons move around the nucleus. These tiny bits carry electric charge and have much less mass than protons or neutrons.



Big rubbed a balloon on a wool sweater, then held it near paper bits. A few bits lifted. Rubbing can move some electrons from the wool onto the balloon. Electrons carry negative charge. The balloon now had extra electrons; the wool had fewer. The tiny bits had moved, not disappeared.



“Charge can be positive or negative,” said Big. Opposite charges pull toward each other. Like charges push apart. Nearby paper can be pulled even when its charges balance overall: the balloon nudges charges inside the paper a little out of place. That makes the nearer side pull toward it.

electron	a tiny bit of matter that carries negative electric charge
charge	a property of matter that can cause electric pulls and pushes
atom	a tiny building block of matter
nucleus	the middle of an atom, holding protons and usually neutrons
wire	a thin metal path that electric current can travel through
static cling	fabrics sticking together because of electric charge

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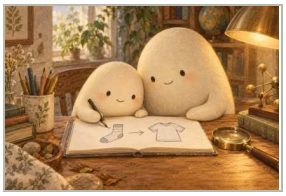
Name _____ Date _____



Next, Big closed a switch on a battery kit. The lamp lit. A wire is a thin metal path. Some electrons in metal can move along it. The closed circuit made a complete loop for electric current. The battery supplied energy; electrons were already in the wires. The lamp did not use them up.



A sock stayed on a shirt when Little lifted it. “Static cling!” said Big. That means fabrics stick because of electric charge. As some fabrics rub and separate, electrons can move between them. The fabrics can end up with opposite charges. Electrons can also jump across a tiny gap as charge spreads out.



Little drew the sock and shirt, with an arrow for electrons moving between them. “The tiny bits were already there,” Little said. “Some changed places.” Big nodded. Little kept one question: “Where do the tiny bits go when a sock stops sticking?”

1. Where do the balloon’s extra electrons come from?
2. How can a sock and shirt end up with opposite charges?
3. How are electrons in a wire like the electrons involved in static cling?

Your turn

Write one command sentence for the balloon test.

Example: Rub the balloon on your sweater, then hold it near the paper bits.